Gate 09 — Husbandry Frontend Route + Profile Teaser

Goal

Ship the public `/species/[id]/husbandry/` route and the conditional teaser block on `/species/[id]/`. This is the user-visible half of the husbandry feature and the artifact that will be demoed at the ECA Workshop on 2026-06-01.

Stories

- **Story 09.1** — As a Tier 1 public user on a species profile whose `has_husbandry` is true, I want a "Keeping this species" teaser block with a link to `/species/[id]/husbandry/`, so that I can discover husbandry guidance without clutter (AC-1).
- **Story 09.2** — As a Tier 1 public user on a species profile whose `has_husbandry` is false, I want **no** husbandry teaser or empty state on the profile, so that under-documented species stay clean (AC-2).
- **Story 09.3** — As a Tier 1 public user, I want a dedicated route `/species/[id]/husbandry/` that renders all structured fields, narrative, references, contributors, last-reviewed-by + date, the sourcing-ethics block, and a platform disclaimer (AC-3).
- **Story 09.4** — As a Tier 1 public user viewing a husbandry page whose review is older than 24 months, I want a subtle "review pending" note near the reviewed-by stamp, so that I can judge content freshness without alarm (AC-4).
- **Story 09.5** — As a Tier 1 public user, I want difficulty presented as **factors** ("water parameter demand: demanding — stable soft water required"; "breeding complexity: requires conditioning + trigger") rather than a single label, so that I draw my own conclusion (locked Q2, 2026-04-18).
- **Story 09.6** — As a Tier 1 public user on a species whose `cares_status` is populated or `shoal_priority` is true, I want the profile's husbandry teaser (when present) visually emphasized as high-interest for breeders, so that CARES/SHOAL priority species are obvious to the hobbyist audience (AC-7).
- **Story 09.7** — As a Tier 1 public user on the husbandry page, I want a "Contribute updates" CTA that links to the contribute form (Gate 10) with species context pre-filled, so that I can flag corrections or offer additional content (AC-6).
- **Story 09.8** — As a Tier 1 public user, I want the husbandry page to link back to the species' linked field programs and (one-way, public-tier view) to the ex-situ aggregate, so that the page is embedded in the rest of the species context (BA Cross-Feature Impact).

Scope Assessment

Story	Frontend	Backend	Full-Stack	Complexity
09.1	✓			S
09.2	✓			S
09.3	✓			L
09.4	✓			S
09.5	✓			M
09.6	✓			S

09.7	✓			S
09.8	✓			S

Frontend-only gate. No backend changes (all contract work landed in Gate 08).

Components and Files

- **New page** `frontend/app/species/[id]/husbandry/page.tsx`.
- **New fetch helper** in `frontend/lib/speciesDetail.ts` (or a new `frontend/lib/husbandry.ts`): `fetchSpeciesHusbandry(id)` returning `{ kind: "ok", data } | { kind: "not_found" } | { kind: "error" }`, mirroring the existing `fetchSpeciesDetail` shape.
- **New component** `frontend/components/HusbandryTeaser.tsx` — renders on the profile only when `sp.has_husbandry` is truthy; visually emphasized when `sp.cares_status` or `sp.shoal_priority`.
- **New component** `frontend/components/HusbandryDifficultyFactors.tsx` — renders the seven `difficulty_*` factor fields as a descriptive list, with blank factors elided.
- **New component** `frontend/components/HusbandryDisclaimer.tsx` — the platform disclaimer (BA §5): "Guidance reflects practices reported by keepers and published sources. Not a protocol. Conditions vary; consult a qualified aquatic veterinarian for health issues."
- **Reuse** `frontend/components/HusbandrySourcingEthics.tsx` — wire into the husbandry page, unchanged.
- **Edit** `frontend/app/species/[id]/page.tsx` — insert `<HusbandryTeaser>` between the Captive Population Summary and the Field Programs sections when `sp.has_husbandry` is true; render nothing otherwise.

Page Layout (`/species/[id]/husbandry/`)

Order top-to-bottom:

1. Breadcrumb: `← {Species italic name}` linking back to `/species/[id]/`.
2. Page title: "Keeping *{Species name}*" (h1).
3. Disclaimer (`HusbandryDisclaimer`).
4. At-a-glance panel: spawning mode, flow preference, min tank volume, sex ratio, live-food required (boolean), CARES registered breeders (boolean). Elide fields that are blank.
5. Difficulty Factors (`HusbandryDifficultyFactors`).
6. Water Parameters (temp / pH / hardness / flow / notes).
7. Tank & System (volume / footprint / aquascape / substrate / cover / notes).
8. Diet (accepted foods list / live-food flag / frequency / notes).
9. Behavior & Social Structure.
10. Breeding (spawning mode / triggers / egg count / fry care / bottlenecks / notes).
11. Narrative (plain text, rendered with `whitespace-pre-line` — paragraphs split on blank lines; no Markdown at MVP per 2026-04-18 decision).
12. Sourcing Ethics (`HusbandrySourcingEthics`) + species-specific sourcing notes from `sourcing_notes`.
13. References (list of `sources`, each a `label` plus optional outbound link).
14. Governance footer: "Reviewed by {reviewer username + ORCID if present} on {last_reviewed_at}." If `review_is_stale`, append in smaller muted text: "Review pending — content may be out of date."
15. Contributors line (when populated).

16. "Contribute updates" CTA — links to Gate 10's route (e.g. `/contribute/husbandry?species={id}`) with species id as a query param.
17. Cross-links: "Field programs for this species →" and "Held at N institutions" (public tier summary — reuse existing species detail data; do not duplicate the ex-situ summary inline, just link back).

Empty / sparse sections: elide entire subsection headings when all their fields are blank. A record may legitimately publish with only narrative + sources; the page must render cleanly in that case.

Acceptance Criteria

AC-09.1 (refines BA AC-1) — Teaser renders only when content exists

Given species X has `has_husbandry: true` on the species detail API **When** any user loads `/species/[id=X]/` **Then** a "Keeping this species" teaser block renders, after the captive population summary and before the field programs section, with link text such as "See husbandry guidance →" pointing to `/species/[id=X]/husbandry/` .

AC-09.2 (refines BA AC-2) — No teaser when content is absent

Given species X has `has_husbandry: false` **When** any user loads `/species/[id=X]/` **Then** no teaser block, empty state, or "coming soon" affordance appears for husbandry. The page renders identically to its pre-Gate-09 layout.

AC-09.3 (refines BA AC-3) — Full husbandry page for Tier 1

Given species X has a published husbandry record with all sections populated **When** an anonymous user loads `/species/[id=X]/husbandry/` **Then** the page renders (in order): page title, disclaimer, at-a-glance panel, difficulty factors, water, tank, diet, behavior, breeding, narrative, sourcing-ethics block, species-specific sourcing notes, references list, reviewed-by stamp with date, contributors, contribute CTA, and cross-links to field programs and ex-situ summary.

AC-09.4 (refines BA AC-4) — Stale review note

Given the API returns `review_is_stale: true` for species X's husbandry **When** an anonymous user loads `/species/[id=X]/husbandry/` **Then** a subtle "Review pending — content may be out of date." note appears adjacent to the reviewed-by stamp. It is **not** a banner, alert, or error styling — muted text only.

AC-09.5 — Difficulty rendered as factors

Given a husbandry record has `difficulty_water_parameter_demand = "demanding — stable soft water required"` and `difficulty_breeding_complexity = "requires conditioning + trigger"` and other difficulty factors blank **When** the page renders **Then** the Difficulty Factors section shows exactly those two factors as labeled descriptive items, and **no** single "Beginner / Intermediate / Advanced" label appears anywhere on the page or the teaser.

AC-09.6 — Absent sections elide

Given a published husbandry record with only `narrative` and `sources` populated (all structured value fields blank) **When** the page renders **Then** the Water, Tank, Diet, Behavior, Breeding, and Difficulty Factors section headings are **not** rendered — the page shows title, disclaimer, narrative, sourcing ethics, references, reviewed-by stamp, and the contribute CTA.

AC-09.7 (refines BA AC-6) — Contribute CTA pre-fills species

Given a user lands on `/species/[id=X]/husbandry/` **When** they click "Contribute updates" **Then** they are navigated to the Gate 10 contribute route with species X identified in the URL (e.g. `?species={X}`), such that the Gate 10 form loads with the species context already selected.

AC-09.8 (refines BA AC-7) — CARES / SHOAL emphasis on teaser

Given species X has `has_husbandry: true` **and** (`cares_status` is populated **or** `shoal_priority` is true) **When** the profile renders **Then** the husbandry teaser has a distinct visual treatment (e.g. accented border or priority badge) indicating the species is of interest to breeders.

And Given species Y has `has_husbandry: true` but is neither CARES nor SHOAL priority **When** the profile renders **Then** the teaser renders with the standard treatment — no emphasis.

AC-09.9 — 404 handling

Given species X's husbandry endpoint returns 404 (no published record) **When** a user manually navigates to `/species/[id=X]/husbandry/` **Then** the route renders a `notFound()` (Next.js 404) rather than an empty shell or a fallback "no content" page. Users arrive at the husbandry route only via the teaser, which only renders when content exists; a manual visit to a non-existent husbandry page is a legitimate 404.

AC-09.10 — Deep-linking and SEO

Given a published husbandry page **When** a crawler or social share card requests the URL **Then** `generateMetadata` returns a title of the form "Keeping *{Species name}* — Madagascar Freshwater Fish" and a description derived from the narrative (first ~160 chars, collapsed whitespace).

AC-09.11 — Sourcing-ethics component is wired unchanged

Given the existing `frontend/components/HusbandrySourcingEthics.tsx` **When** the husbandry page renders **Then** the component is used as-is (no prop changes, no fork); its copy is the single source of truth for the ethics block.

Out of Scope

- Gate 10's form implementation itself (this gate links to the route; Gate 10 ships the form).
- Photos, image galleries, or tank setup visuals (post-MVP per BA §6).
- "Show me beginner-appropriate CARES species" filter on the directory (post-MVP; the factor data exists but filter UI is deferred).
- Version history, diffs, comments, ratings (post-MVP).
- Any Tier 2+ write surface (post-MVP contribution pipeline).
- Client-side mutation — this gate is render-only.

Dependencies

- Gate 08 API endpoints live in staging.
- At least one seeded `SpeciesHusbandry` record for visual verification. If Aleksei hasn't authored one yet, use the fixture from Gate 08 AC-8.1.
- `fetchSpeciesDetail` continues to expose `has_husbandry` , `cares_status` , `shoal_priority` , `ex_situ_summary` , `field_programs` — already contracted in Gate 08.

Sequencing / Deadline Note (June 1, 2026)

This gate is what the workshop demo shows. Target merge: **2026-05-22** to leave 10 days for Aleksei to author final exemplars on the live system, warm-cache the exemplar URLs, and do a run-through with the

prod deploy pipeline (warm-cache.sh already covers species profile URLs; extend to cover husbandry URLs as part of this gate — small tweak to `warm-cache.sh`).

If Gate 10 (contribute form) slips, this gate can still ship with the "Contribute updates" CTA pointing to a placeholder route that renders "Contribute flow coming soon — email alex.saunders@wildlifeprotectionsolutions.org" as a temporary fallback. Demo does not require the contribute form to be functional; it requires it to be **present** as a visible frame.

Test Writer Guidance

At this gate, the test writer should verify:

- Teaser shows/hides correctly based on `has_husbandry` (both directions).
- CARES / SHOAL emphasis flips on the teaser when priority flags change.
- Husbandry page renders full content with all sections; renders gracefully (section elision) when sections are empty.
- "Review pending" note appears iff `review_is_stale` is true. Boundary: 24 months minus 1 day = not stale; 24 months plus 1 day = stale.
- "Contribute updates" link carries species id in the URL.
- No single difficulty label appears anywhere in the DOM (assert against "Beginner", "Intermediate", "Advanced", "Expert-only" strings in the rendered page).
- **Adversarial:** manually visiting `/species/{unpublished-id}/husbandry/` returns Next.js 404, not a shell with `notFound` content leaked.
- **Adversarial:** species with `has_husbandry: false` must not render the teaser even if the user cached a stale page — verify via server-rendered HTML, not client-side hydration.
- **Accessibility:** disclaimer and sourcing-ethics blocks are reachable by screen reader in reading order; headings form a valid h1 → h2 hierarchy with no skipped levels; all outbound reference links have discernible names.
- **SEO:** `<title>` includes the species' scientific name; meta description is present and non-empty.

Risks and Open Questions

- **Authoring capacity (Q1, still open).** Demo quality depends on 3–5 rich exemplars. If only 1 lands, the demo still works for one species; the "coming soon" framing for the rest is a feature (per BA §6).
- **Visual design of emphasized teaser. Resolved 2026-04-18** by ux-reviewer — see "Visual treatment — teaser block" section below.
- **Markdown rendering in narrative. Resolved 2026-04-18:** plain text at MVP. Matches existing long-text convention (`Species.description`, `ConservationAssessment.rationale` — all plain `TextField` rendered as `<p>{text}</p>`). No `dangerouslySetInnerHTML`, no sanitizer dependency, zero XSS surface at Tier 1. Render with Tailwind `whitespace-pre-line` so `\n\n` paragraph breaks survive. Re-evaluate when Tier 3+ contributor submissions land post-MVP (likely a restricted-subset `react-markdown` + `rehype-sanitize` at that point).

Visual treatment — teaser block

Locked 2026-04-18 by ux-reviewer. Resolves the "Visual design of emphasized teaser" item above.

Emphasized variant (CARES-listed OR `shoal_priority`): left accent border, 3–4px, in a cool editorial tone (recommend `border-l-sky-600` or `border-l-emerald-600` — stay out of the red/orange/amber band owned by the IUCN badge), plus a small uppercase chip label reading `CARES breeder priority` or `SHOAL priority` rendered inline above the "Keeping this species" h2. The rest of the block — heading, 1–

2 line teaser sentence, "See husbandry guidance →" link — keeps default profile typography. No background fill, no shadow, no icon. Reads as editorial pull-quote, not promotion.

Justification: the profile header already carries a saturated IUCN status pill. A tinted background or second colored badge near it would create badge-on-badge noise and muddy the conservation-first hierarchy the BA locked. A left accent border is a long-established content convention (callouts, pull-quotes) that signals "this is relevant to you" without competing chroma. The chip is text-first so it survives grayscale, print, and RSS.

Placement: between Captive Population Summary and Field Programs (as already specified in Components and Files). Matches the hobbyist JTBD ("can I keep this fish?") arriving after the reader has seen conservation status, ecology, and whether any institutions already hold it — which is the context they need to evaluate "should I take this on?" Above ex-situ would front-run the conservation framing; below field programs would bury it.

Normal (non-emphasized) variant: same block, same copy, **no** accent border, **no** chip. Plain heading + teaser sentence + link. Emphasized variant is strictly additive — one border, one chip — so the two variants share a component and the CARES/SHOAL case is a prop-driven overlay, not a parallel design.

Accessibility:

- Do not rely on color alone. Chip carries the text label `CARES breeder priority` / `SHOAL priority` ; accent border is decorative and must be paired with that chip so color-blind and grayscale users get the signal.
- Chip contrast: text on chip must meet WCAG AA (4.5:1) at small-text size.
- Accent border is `aria-hidden` decoration; no role change.
- Teaser block wraps heading and link in `<section aria-labelledby="husbandry-teaser-heading">` so screen-reader users land on "Keeping this species" as a landmark.
- When emphasized, chip text must appear in reading order before the heading so a screen-reader user hears "CARES breeder priority. Keeping this species. See husbandry guidance." in that order.
- Link text "See husbandry guidance →" is self-describing; the arrow is `aria-hidden` .
- If both CARES and SHOAL flags are set, show a single chip reading `CARES
  - SHOAL priority` rather than stacking two chips.

Deliberately avoided: background tints (compete with sparse-data amber warning), icons (read as marketing), size/weight changes on the heading (break h2 rhythm shared with Captive Population Summary and Field Programs).